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UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA
Scholarship with Professionalism

BCPC 112: INTRODUCTION TO BUSINESS STATISTICS

LECTURE NINE

Correlation Analysis

Learning Objectives

In this lecture, you learn to:

- Explain the meaning and purpose of correlation analysis.
- Distinguish between correlation and regression.
- Construct and interpret scatter plots.
- Describe the direction and strength of relationships.
- Compute and interpret Pearson's and Spearman's correlation coefficients.
- Compute the correlation coefficient

Correlation vs. Regression

- Correlation analysis is a statistical method used to measure the association between variables.
- Regarding the association, Correlation:
  - Indicates strength.
  - Indicates direction.
  - Does not imply causation.

Correlation vs. Regression

- Regression analysis is used to:
  - describe and estimate the nature of the relationship between variables—that is, positive or negative, linear or nonlinear.
  - Predict the value of a dependent variable based on the value of at least one independent variable
  - Explain the impact/effect of changes in an independent variable on the dependent variable

Dependent variable: the variable we wish to predict or explain

Independent variable: the variable used to predict or explain the dependent variable

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Correlation vs. Regression

- The purpose of this topic is to answer these questions statistically:

1. Are two variables related?- is there a particular observable or logical relationship between two variables? If one variable increases, what happens to the other variable? Does it increase or decrease?
    - Consider sales and revenue; motivation and worker performance; exchange rate and imports; smoking and lung cancer; weight and blood pressure; interest rate and investment; study time and score.

Correlation vs. Regression

2. If there is a relationship, what type of relationship exists?
   - Is it positive or negative?
3. What is the strength of the linear relationship?
   - Is it a weak relationship or a strong relationship?
4. What kind of predictions can be made from the relationship?

Correlation vs. Regression

- To answer question 1,2 and 3,
  - the correlation coefficient, a numerical measure to determine whether two variables are related and to determine the strength of the relationship between the variables is used.
- To answer question 4,
  - regression will be used.

Scatter Plots and Correlation

- The first step in correlation is to have a conceptual or logical understanding of the relation between the two variables.
- The next step is to construct a scatter plot for the data to provide a visual indication of whether a relationship exists before numerical analysis is performed.
- A **scatter plot** is a graph of the ordered pairs (x, y), where one variable is plotted on the horizontal axis and the other on the vertical axis.
- Either variable may be placed on either axis because correlation is symmetric.

Example1: Car Rental Companies

Construct a scatter plot for the data shown for car rental companies in the United States for a recent year.

Company	Cars (in ten thousands)	Revenue (in billions)
A	63.0	$7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

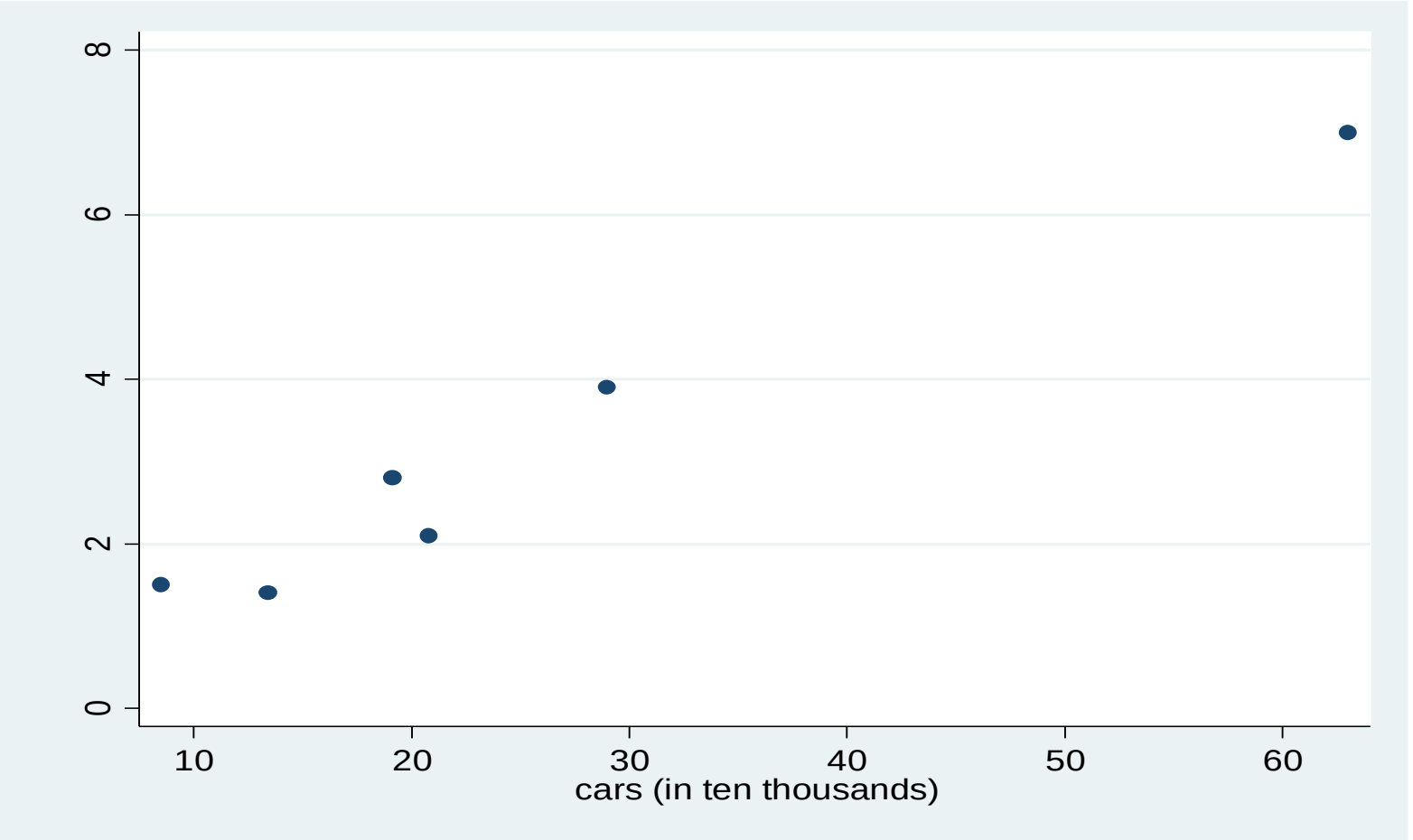
Step 1: Draw and label the *x* and *y* axes and denote one of the variables by X and the other by Y.

Step 2: Plot each point on the graph.

Example1: Car Rental Companies

- What do you think is the conceptual or logical relationship between the number of cars given out to hire and the sales revenue made?
  - Positive? Negative? No relationship?
- As the number of cars available for hire increases, does the sales revenue of the firm decrease or increase?
- The scatter plot is shown in the following diagram

Example1: Car Rental Companies



Example 2: Absences/Final Grades

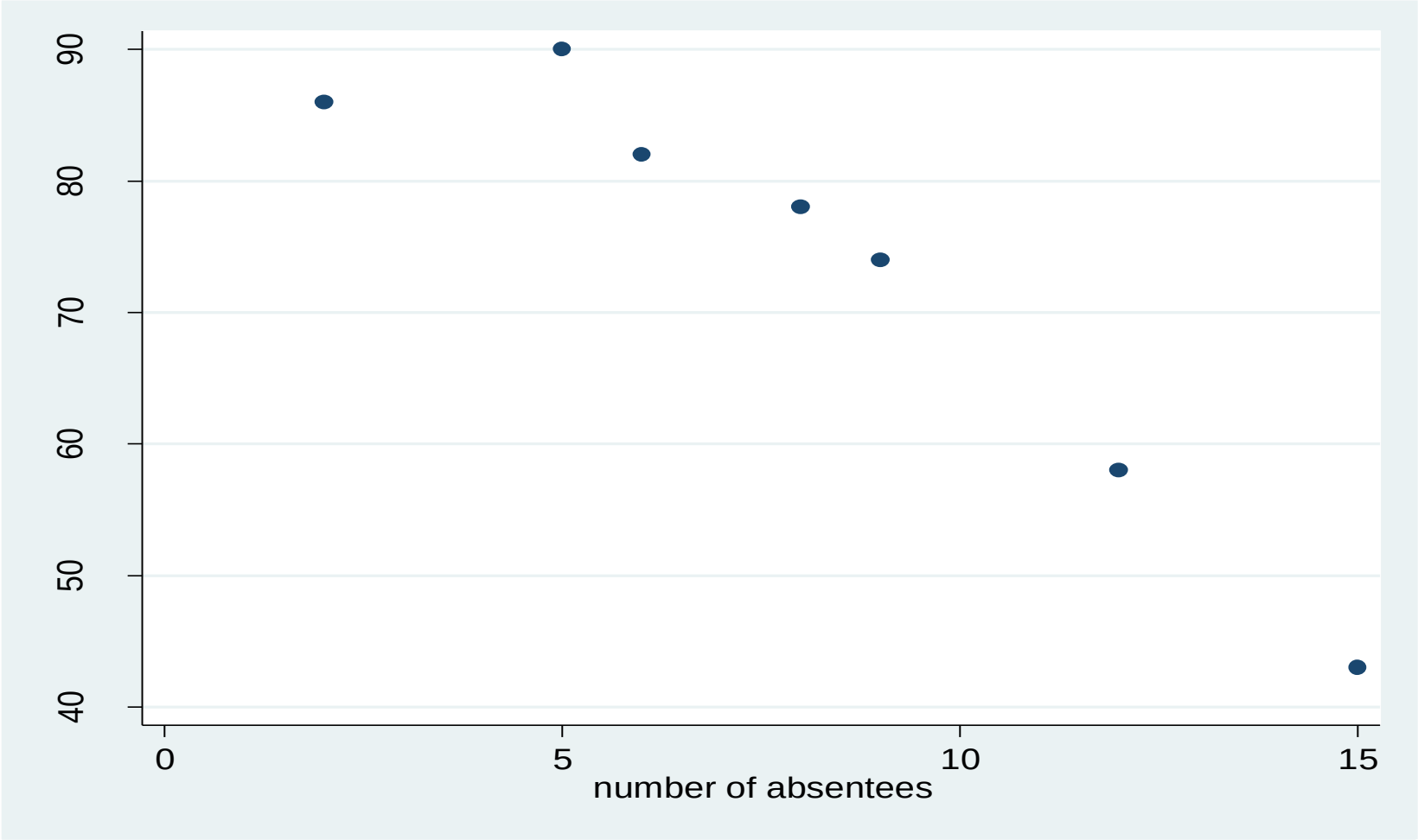
Construct a scatter plot for the data obtained in a study on the number of absences and the final grades of seven randomly selected students from a statistics class.

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

Step 1: Draw and label the *x* and *y* axes and denote one of the variables by X and the other by Y

Step 2: Plot each point on the graph.

Example 2: Absences/Final Grades



Correlation-interpretation

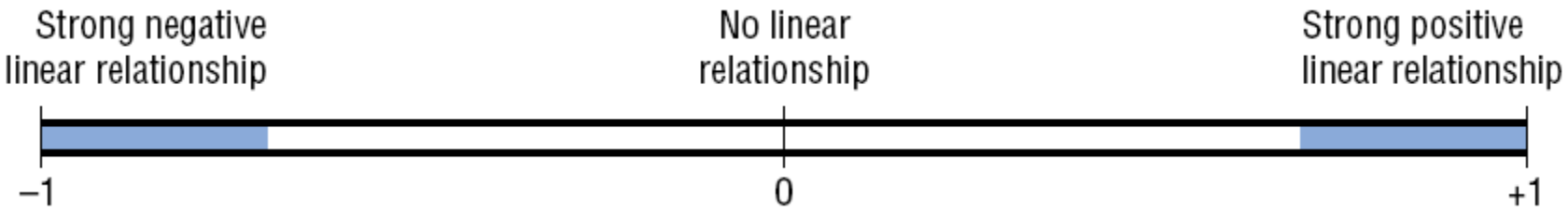
- After the scatter plot, the next step is to numerically calculate the correlation coefficient between the variables and interpret the coefficient
- The **correlation coefficient** computed from the sample data measures the strength and direction of a linear relationship between two variables.
- There are several types of correlation coefficients. Two will be explained in this course;
- The two explained in this course are the **Pearson product moment correlation coefficient (PPMC)** and the **Spearman's Rank correlation coefficient**

Correlation-interpretation

- The symbol for the sample correlation coefficient is *r*. The symbol for the population correlation coefficient is ρ.

Correlation-interpretation

- The range of the correlation coefficient is from –1 to +1.
- If there is a **strong positive linear relationship** between the variables, the value of *r* will be close to +1.
- If there is a **strong negative linear relationship** between the variables, the value of *r* will be close to –1.



Correlation-interpretation

- If $r=1$, then there exists a perfect positive relationship between the variables
- If $r=-1$, then there exists a perfect negative relationship between the variables
- If $r=0$, then there exists no relationship between the variables (no correlation)
- If $0<r<0.5$, then there exists a weak positive relationship between the variables
- If $0.5<r<1$, then there exists a strong positive relationship between the variables

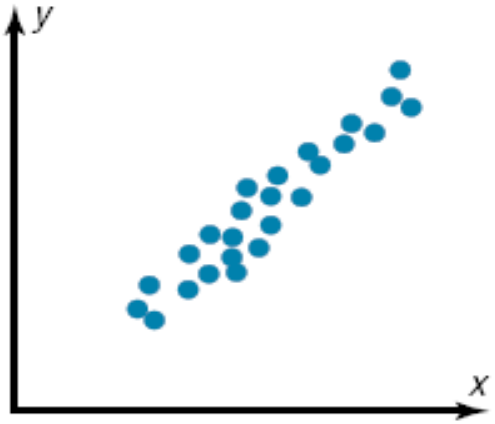
Correlation-interpretation

- If -0.5<*r<0,* then there exists a weak negative relationship between the variables
- If -0.5<*r<-1,* then there exists a strong negative relationship between the variables

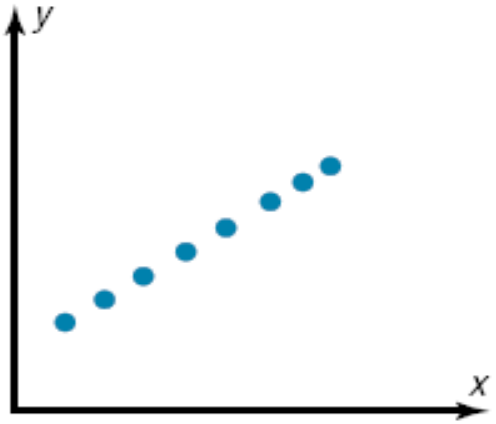
Correlation-interpretation



(a) $r = 0.50$



(b) $r = 0.90$



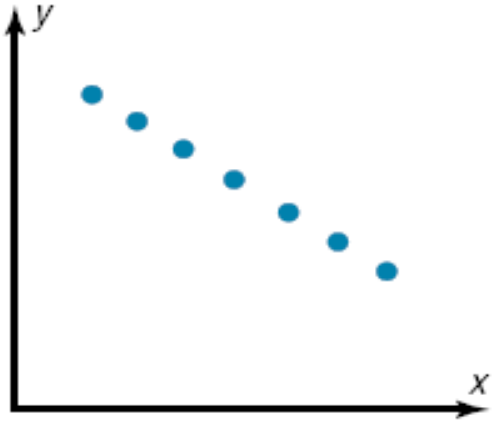
(c) $r = 1.00$



(d) $r = -0.50$



(e) $r = -0.90$



(f) $r = -1.00$

Correlation Coefficient (PPMC)

The Pearson's correlation coefficient (r) is the most widely used measure of the strength and direction of a linear relationship between two quantitative variables.

$$r = \frac{n\left(\sum xy\right) - \left(\sum x\right)\left(\sum y\right)}{\sqrt{\left[n\left(\sum x^2\right) - \left(\sum x\right)^2\right]\left[n\left(\sum y^2\right) - \left(\sum y\right)^2\right]}}$$

where *n* is the number of data pairs.

Correlation Coefficient (PPMCC)

- It can also be expressed in notation form as:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}$$

- Where;

$$S_{xy} = n\left(\sum xy\right) - \left(\sum x\right)\left(\sum y\right)$$

$$S_{yy} = \left[n\left(\sum y^2\right) - \left(\sum y\right)^2\right]$$

Example 1: Car Rental Companies

Compute the correlation coefficient for the data in Example 1.

Company	Cars x (in 10,000s)	Income y (in billions)	xy	x^2	y^2
A	63.0	7.0	441.00	3969.00	49.00
B	29.0	3.9	113.10	841.00	15.21
C	20.8	2.1	43.68	432.64	4.41
D	19.1	2.8	53.48	364.81	7.84
E	13.4	1.4	18.76	179.56	1.96
F	8.5	1.5	12.75	72.25	2.25
	$\Sigma x =$ 153.8	$\Sigma y =$ 18.7	$\Sigma xy =$ 682.77	$\Sigma x^2 =$ 5859.26	$\Sigma y^2 =$ 80.67

Example 1: Car Rental Companies

Compute the correlation coefficient for the data in Example 1.

$\Sigma x = 153.8$, $\Sigma y = 18.7$, $\Sigma xy = 682.77$, $\Sigma x^2 = 5859.26$, $\Sigma y^2 = 80.67$, $n = 6$

$$r = \frac{n\left(\sum xy\right) - \left(\sum x\right)\left(\sum y\right)}{\sqrt{\left[n\left(\sum x^2\right) - \left(\sum x\right)^2\right]\left[n\left(\sum y^2\right) - \left(\sum y\right)^2\right]}}$$

$$r = \frac{(6)(682.77) - (153.8)(18.7)}{\sqrt{\left[(6)(5859.26) - (153.8)^2\right]\left[(6)(80.67) - (18.7)^2\right]}}$$

$r = 0.982$ (strong positive relationship)

Example 2: Absences/Final Grades

Compute the correlation coefficient for the data in Example 2.

Student	Number of absences, x	Final Grade y (pct.)	xy	x^2	y^2
A	6	82	492	36	6,724
B	2	86	172	4	7,396
C	15	43	645	225	1,849
D	9	74	666	81	5,476
E	12	58	696	144	3,364
F	5	90	450	25	8,100
G	8	78	624	64	6,084
	$\Sigma x =$ 57	$\Sigma y =$ 511	$\Sigma xy =$ 3745	$\Sigma x^2 =$ 579	$\Sigma y^2 =$ 38,993

Example 2: Absences/Final Grades

Compute the correlation coefficient for the data in Example 2.

$\Sigma x = 57$, $\Sigma y = 511$, $\Sigma xy = 3745$, $\Sigma x^2 = 579$,
$\Sigma y^2 = 38{,}993$, $n = 7$

$$r = \frac{n\left(\sum xy\right) - \left(\sum x\right)\left(\sum y\right)}{\sqrt{\left[n\left(\sum x^2\right) - \left(\sum x\right)^2\right]\left[n\left(\sum y^2\right) - \left(\sum y\right)^2\right]}}$$

$$r = \frac{(7)(3745) - (57)(511)}{\sqrt{\left[(7)(579) - (57)^2\right]\left[(7)(38{,}993) - (511)^2\right]}}$$

$r = -0.944$ (strong negative relationship)

Spearman's Rank Correlation Coefficient

- The formulas is given by:

$$r_s = 1 - \frac{6\left(\sum d^2\right)}{n(n^2 - 1)}$$

- Where *n* is the number of observations. *d* is the difference in ranks between X and Y.
- The interpretation for the spearman's correlation is the same as for the PPMCC

Step by Step Approach

1. Rank X and Y in an ascending or descending order
2. Calculate the difference between the ranks
3. Find the square of the difference and calculate

Example-Step by Step Approach

- Consider the data from Example 1

COMPANY	CARS(in 10,000s)	Income (in billions)
A	63	7
B	29	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

Example-Step by Step Approach

- Represent one of the variables by X and the other by Y. Remember that correlation is symmetric
- Order the data in ascending order
- Compute the difference between the ranks
- Calculate the correlation coefficient

Example-Step by Step Approach

Company	Cars(X)	Income (Y)	Rank of X	Rank of Y	d	d^2
A	63	7	6	6	0	0
B	29	3.9	5	5	0	0
C	20.8	2.1	4	3	1	1
D	19.1	2.8	3	4	-1	1
E	13.4	1.4	2	1	1	1
F	8.5	1.5	1	2	-1	1
Total						**4**

Example-Step by Step Approach

- The spearman's rank correlation coefficient is given as:

$$r_s = 1 - \frac{6(4)}{6(6^2 - 1)} = 0.8857$$

- Note: the answer will not necessarily be the same but the interpretation will be the same. In this case, there exists a strong positive correlation between car rentals and revenue

Special cases

- How do we rank the data when two or more data points have the same rank?
- When two scores tie in rank, both are given the mean of the two ranks they would occupy and the next rank is eliminated to keep ***n*** (the number of observations) consistent. For example, if two data points tied for 4th place, both would receive a rank of 4.5 ((4 + 5) ÷ 2), and the next data point would be ranked number 6.
- If three points tied for 4th place, the three would receive a rank of 5 ((4 + 5+6) ÷ 3), and the next school would be ranked number 7.

Example-Special cases

- Consider the following example

Company	Cars(X)	Income (Y)	Rank of X	Rank of Y	d	d^2
A	63	7	6	6	0	0
B	29	2.1	5	4	1	1
C	13.4	2.1	2.5	4	-1.5	2.25
D	19.1	2.1	4	4	0	0
E	13.4	1.4	2.5	1	1.5	2.25
F	8.5	1.5	1	2	-1	1
Total						**6.5**

Example-Special cases

- The spearman's rank correlation coefficient is given as:

$$r_s = 1 - \frac{6(6.5)}{6(6^2 - 1)} = 0.81$$

- This indicates a strong positive correlation between the two variables.

Exercise 9.1

- It is believed that there is a correlation between retirement age and life expectancy. As a result, a research firm collected data on average retirement age and average life expectancy for various communities in the region. The information is depicted below:

Community	A	B	C	D	E	F	G
Avg. retirement age	40	61	55	62	48	45	60
Avg. life expectancy	56	70	65	80	60	55	78

- Calculate the correlation coefficient using the Pearson product moment and interpret your result

Exercise 9.2

- The table below summarizes the linear relation between predicted positions for athletes before the event and the actual positions after the event.

Athlete	A	B	C	D	E	F	G
Predict Position	6	3	7	1	5	2	4
Actual Position	2	7	3	4	1	6	5

- Use the spearman's rank to calculate the correlation coefficient and interpret your result.

THANK YOU!